

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL AS

FURTHER MATHEMATICS

(9665/FM02) Unit FPSM1 Pure Mathematics, Statistics and Mechanics

Thursday 16 May 2024

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

- For this paper you must have the OxfordAQA Booklet of Formulae and Statistical Tables (enclosed).
- You may use a graphical calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- There are three sections to this paper.
- The maximum mark for this paper is 80. There are 40 marks for **Section A**, 20 marks for **Section B** and 20 marks for **Section C**.

Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- Show all necessary working; otherwise marks may be lost.

For Examiner's Use	
Question	Mark
1	
2	
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11	
12	
TOTAL	

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Pure Mathematics

1 A curve passes through the point $(1, -3)$ and satisfies the differential equation

$$\frac{dy}{dx} = \sqrt{x^2 + 2y^2}$$

Give your answer to four decimal places.

[5 marks]

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Answer

5

Turn over ►



2 The matrix **P** is defined by $\mathbf{P} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ where k is a positive constant.

The matrix **Q** is defined by $\mathbf{Q} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$

2 (a) Describe the transformation represented by matrix **P**

[1 mark]

2 (b) (i) Calculate the matrix **QP** giving your answer in terms of k

[2 marks]

Answer _____



2 (b) (ii) A triangle T_1 has an area of 4.5 square units.

The triangle T_2 is the image of T_1 under the transformation represented by the matrix **QP**

Find the area of triangle T_2

[3 marks]

Answer _____

6

Turn over for the next question

Turn over ►



3

$$f(x) = 8x^3 - 28x^2 + 34x - 17$$

The equation $f(x)=0$ has one positive root α

3 (a)

[2 marks]

3 (b)

[4 marks]

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Answer

$$v = \frac{u}{a + bu} \quad \text{for } u \neq 0 \text{ and } v \neq 0$$

4 (a) It is given that $V = \frac{1}{v}$ and $U = \frac{1}{u}$

[2 marks]

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- 4 (b)** The table below shows some values of u and v obtained in an experiment.

u	1	2	5	10	50
v	0.117	0.167	0.230	0.251	0.272

- 4 (b) (i)** Complete the table below.

Give your values of V to one decimal place.

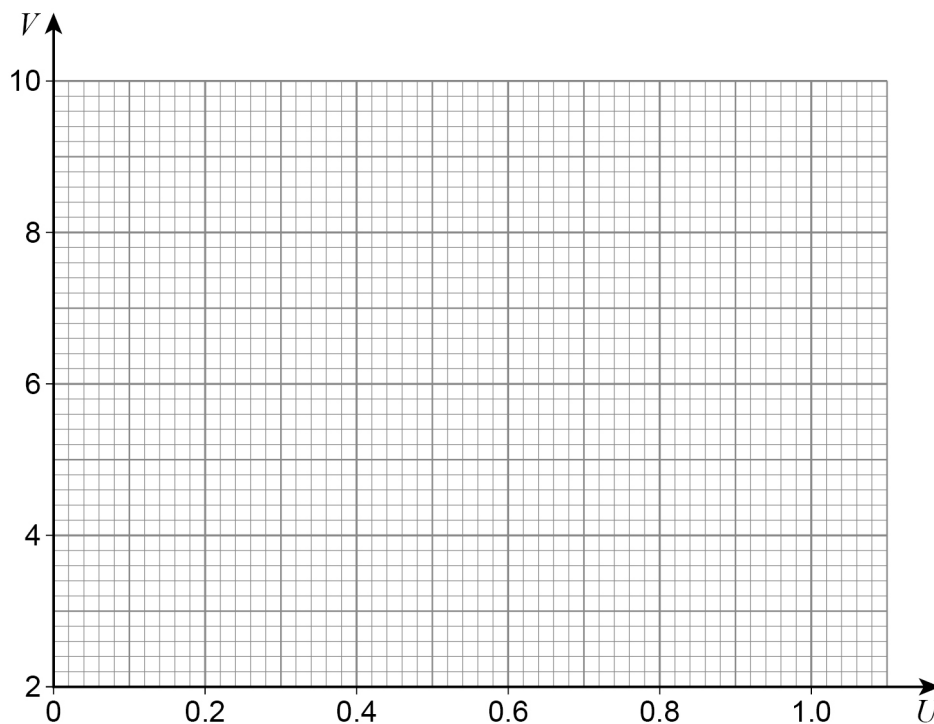
[2 marks]

U	1	0.5	0.2	0.1	0.02
V	8.5				

- 4 (b) (ii)** On the grid plot V against U

Draw a line of best fit for the points you have plotted.

[2 marks]



- 4 (c)** Use your line of best fit to estimate the value of a and the value of b

Give your values to one decimal place.

[2 marks]

$a =$ _____

$b =$ _____

- 4 (d)** Use your values of a and b to estimate the value of u when $v = 0.26$

Give your answer to two significant figures.

[2 marks]

Answer _____

10

Turn over ►



The matrix $\mathbf{A}$ is defined by $\mathbf{A} =$

$$\begin{bmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

Find the possible values of m

Give your answers in exact form.

[4 marks]

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Answer

5 (a) (ii) Hence find the equations of the invariant lines of the transformation represented by **A** **[3 marks]**

Answer



5 (b) The matrix **B** represents a reflection.

The transformation represented by the matrix **AB** is an anticlockwise rotation of $\frac{\pi}{2}$ radians about the origin.

[6 marks]

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Answer



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Section B**Statistics**

Answer **all** questions in the spaces provided.

- 6** The probability that Darius goes for a morning run is 0.35
- The probability that he goes to a cafe given that he goes for a morning run is 0.79
- The probability that he goes to a cafe given that he does **not** go for a morning run is 0.42
- Find the probability that he does **not** go for a morning run given that he does **not** go to a cafe.
- [3 marks]**

Answer _____

3

Turn over for the next question

Turn over ►



- 7** In an experiment, a biased coin is tossed 1000 times and lands with heads facing upwards 653 times.

- 7 (a)** Write down an estimate for the probability that the coin lands with heads facing upwards when the coin is tossed once.

[1 mark]

Answer _____

- 7 (b)** The random variable H is defined as

$$H = \begin{cases} 1 & \text{coin lands with heads facing upwards} \\ 0 & \text{otherwise} \end{cases}$$

when the coin is tossed once.

The probability generating function of H is $G_H(t)$

- 7 (b) (i)** Use your answer to **part (a)** to find $G_H(t)$

[1 mark]

Answer _____



[4 marks]

[illegible]

Answer _____

6

Turn over ►



It is given that $P(X \leq 3) = \frac{61}{125}$

[2 marks]

Answer _____

It is given that $P(Y > 4) = \frac{7}{66}E(Y)$

[4 marks]

[illegible]

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Section C**Mechanics**Answer **all** questions in the spaces provided.

- 9** Two model boats A and B are moving on a lake.

The velocity of A is $\begin{bmatrix} 2.5 \\ -1 \end{bmatrix} \text{ m s}^{-1}$

The velocity of B is $\begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix} \text{ m s}^{-1}$

At time $t = 0$ the displacement of B relative to A is $\begin{bmatrix} 20 \\ d \end{bmatrix}$ metres where d is a constant.

- 9 (a)** Find the velocity of B relative to A

[2 marks]

Answer _____

- 9 (b)** Hence find the value of d given that the two boats collide.

[3 marks]

Answer _____



Show that the dimensions of pressure are $ML^{-1}T^{-2}$

$$P = \rho gh$$

ρ is the density of the liquid in kg m^{-3}

g is the acceleration due to gravity in m s^{-2}

Show that the formula $P = \rho gh$ is dimensionally consistent.

[illegible]

4



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- 11** Two particles P and Q are moving towards each other along a straight line on a smooth horizontal surface.

The particles collide.

Particle P has mass 5 kg and before the collision its speed is 3 m s^{-1}

Particle Q has mass 3 kg and before the collision its speed is 2 m s^{-1}

After the collision the direction of motion of particle P is unchanged and it moves with speed 0.375 m s^{-1}

- 11 (a)** Find the speed of particle Q after the collision.

[3 marks]

Answer _____

- 11 (b)** Find the magnitude of the impulse exerted on particle P by particle Q during the collision.

[2 marks]

Answer _____



- 12** A disc of mass 0.2 kg is sliding on a smooth horizontal surface when it collides with a fixed smooth vertical wall.

The wall is perpendicular to the path of the disc.

The magnitude of the force, F newtons, exerted on the disc by the wall at time t seconds after the disc first makes contact with the wall is given by

$$F = 1800(t - 10t^2)$$

for $0 \leq t \leq 0.1$

The disc is in contact with the wall for 0.1 seconds.

- 12 (a)** Find the magnitude of the impulse exerted on the disc by the wall due to the collision.

State the units of your answer.

[2 marks]

Answer _____ Units _____



Find the range of possible values of u

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Answer

END OF QUESTIONS

6



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